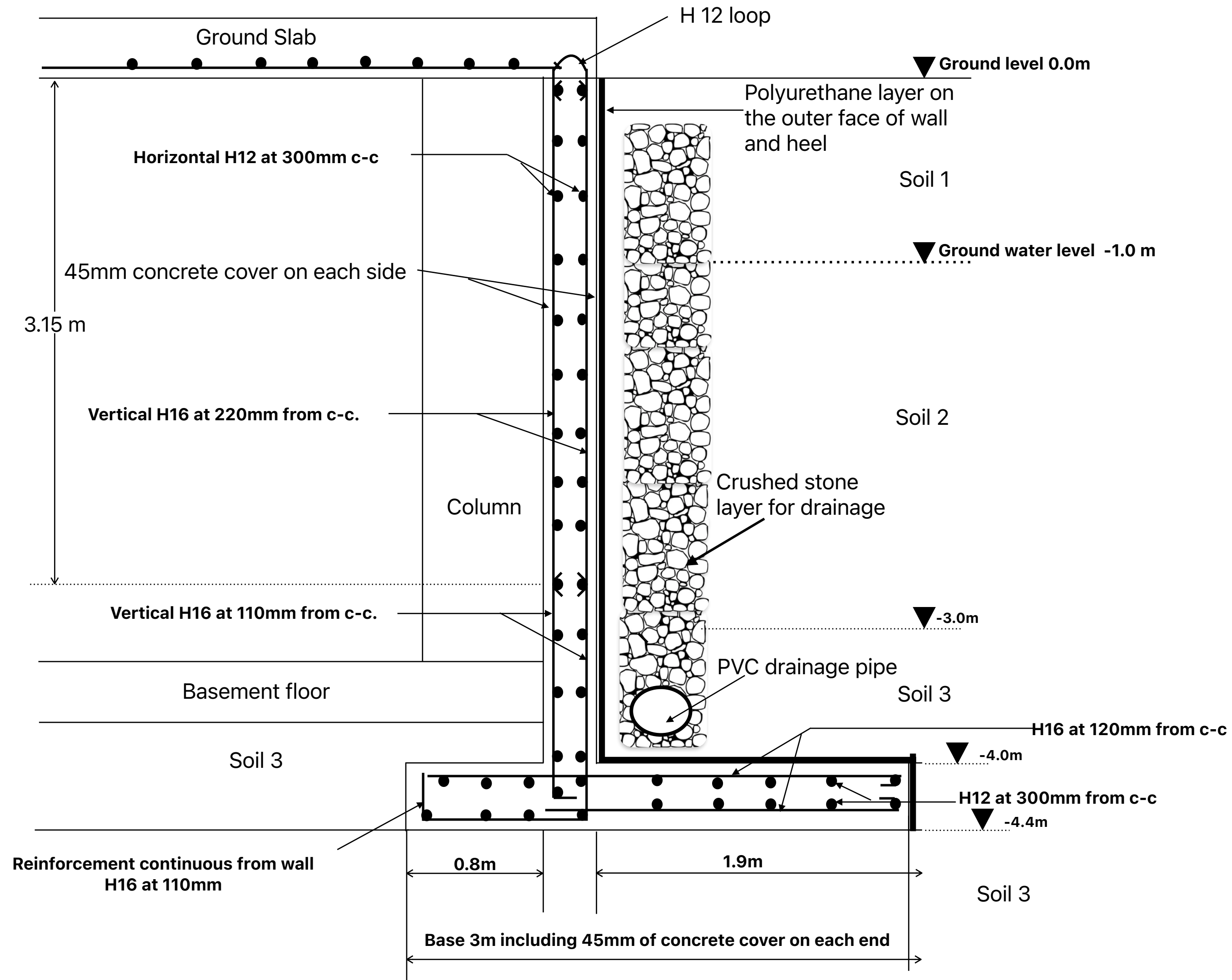


Section FF: (from the Basement plan)



Notes:

- 1- All dimensions for steel reinforcement are in (mm).
- 2- (c-c) means from centre to centre of steel bars.
- 3- The polyurethane layer is to protect from water and chemicals (Sulphates).
- 4- The spacing of vertical reinforcement is to change at depth of 3.15m from ground.

Characteristics of Soil:

- 1- Soil 1: course fill , moist, unit weight of 20 kN/m^3 , friction angle of 32 degrees.
- 2- Soil 2: course fill, saturated, unit weight of 22 kN/m^3 , friction angle of 32 degrees.
- 3- Soil 3: Dense sand and gravel, saturated, unit weight of 20.33 kN/m^3 , friction angle of 40 degrees.
- 4- Soil 4: Same as soil 3 but moist with unit weight of 19.44 kN/m^3 .

Construction:

- 1- Temporary sheet piles are to be driven before excavation.
- 2- Dewatering may be necessary, a single stage well point is suggested.
- 3- Excavation is to be up to 4.4 m below ground.
- 4- The drainage system of course layer and pipe is suggested to prevent ground water to rise.
- 5- The retaining wall is constructed first and a loop of 12mm steel is left above the wall to be tied with the reinforcement of the roof slab.
- 6- The roof slab is to be casted on top of the wall.